

Fujita’s freeness conjecture

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Abstract

We prove Fujita’s freeness conjecture. If X is a smooth complex projective variety of dimension n and L is an ample line bundle, then $K_X + mL$ is globally generated for every integer $m \geq n + 1$.

1 Introduction

We use additive notation for line bundles. Fujita’s freeness conjecture [8] predicts that the canonical bundle becomes globally generated after twisting by $n + 1$ copies of an arbitrary ample line bundle on a smooth projective n -fold. The ample bundle itself need not have any nonzero sections. We prove the following precise form.

Theorem 1.1. *Let X be a smooth connected projective variety over $\mathbb{C}$ of dimension $n \geq 1$, and let L be an ample line bundle on X . Then*

$$K_X + (n + 1)L$$

is generated by its global sections.

The connectedness assumption only fixes notation: the assertion applies to each connected component. In dimension zero every line bundle is globally generated.

The exponent is sharp: for $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(1)$, the bundle $K_X + (n + 1)L$ is trivial, whereas $K_X + nL$ has no nonzero sections. Global generation asks for a nonvanishing section at each point; it does not assert separation of points or tangent directions, or the very-ampleness part of Fujita’s conjecture. Theorem 1.1 also implies that $K_X + mL$ is globally generated for every integer $m \geq n + 1$, even when L is not globally generated; see Corollary 6.3.

History and methods

For curves, Riemann–Roch gives the sharp bound. The surface case follows from Reider’s theorem, and the threefold, fourfold, and fivefold cases were proved by Ein–Lazarsfeld, Kawamata, and Ye–Zhu, respectively [6, 19, 22, 25]. In arbitrary dimension, Angehrn–Siu obtained generation for $m \geq (n^2 + n + 2)/2$ [1]. Helmke developed multiplicity estimates for the centers appearing in the vanishing argument [14, 15]; Heier combined these estimates with the Angehrn–Siu method to obtain a bound of order $n^{4/3}$ [13]. Ghidelli–Lacini subsequently obtained the bound $m \geq n(\log \log n + 2.34)$ for $n \geq 2$ [11, Theorem 1.1]. Han’s recent preprint gives $m \geq \lceil C_0 n \rceil$, where $C_0 = 1.77629988 \dots$ [12, Theorem 1.1]. Chan announced a proof of the stronger intersection-number formulation in [4, Theorem 1.3]; that preprint was withdrawn after the local extension argument did not guarantee the required vanishing order.

The common vanishing strategy constructs controlled singularities at the prescribed point, reduces a suitable center to a point, and then lifts a nonzero fiber value [6, 19]. A recurring difficulty is that a positive-dimensional center can itself be singular, and cutting it down costs positivity. Angehrn–Siu used movement to a smooth point and semicontinuity. Fujita estimated ranks of restriction maps to the exceptional divisor of a point blowup, while Helmke bounded the multiplicities of log canonical centers [1, 9, 14, 15]. Our argument controls a minimizing center by comparing the growth of images of restriction maps with a tangential first variation. The estimate is uniform over all integral subvarieties of a fixed dimension, including singular subvarieties of unbounded degree. The final lift uses the local discrepancy-and-vanishing mechanism of [9, §1(1.6)]; the minimizing construction produces the configuration to which it applies.

The convex viewpoint has a related ancestry. Flag valuations, initial-exponent semigroups, and their relation to the dimensions of section spaces appear in Okounkov’s work [21] and are developed systematically by Lazarsfeld–Mustață and Kaveh–Khovanskii [17, 20]. Boucksom–Chen study the asymptotic distributions of filtration jumps [3], and Blum–Jonsson prove attainment for valuative log-canonical and stability thresholds of ample bundles [2, Theorem E]. The exponential average considered here is a different finite-degree objective, whose attainment is proved in Section 4. The center argument uses elementary counts of initial exponents and the Brunn–Minkowski inequality directly.

The proof

Section 2 fixes the discrepancy normalization and the threshold and vanishing results used below. Fix a point x and suppose that the adjoint bundle is not generated there. An equality-case argument first gives a strict asymptotic inequality for bases of $H^0(X, mL)$ adapted to ordinary vanishing at x : their mean order divided by m has lower limit greater than $n/(n+1)$ (Section 3). We then minimize

$$\frac{1}{h^0(X, mL)} \sum_i \exp\left(\frac{A(v)}{t} - \frac{v(s_i)}{m}\right), \quad t < n+1,$$

over bases (s_i) of $H^0(X, mL)$ and weighted monomial orders v whose closed centers contain x . Here $A(v)$ is the log discrepancy, with the same scaling as v . This functional links three parts of the proof: its derivative under scaling detects the strict ordinary-order excess; its derivative in tangential directions measures a weighted first moment; and its gradient gives positive coefficients for the rational divisor used in the final lift.

Two obstacles enter this approach. First, both bases and models vary. Simultaneous lower bounds on normalized section orders can be encoded by an ideal on the product of X and the complete flag variety of $H^0(X, mL)$. Threshold semicontinuity and retraction to one log resolution then give attainment of the minimum, without compactifying the space of all bases or all valuations (Section 4). Second, a first variation in directions tangent to a positive-dimensional center gives an upper bound for a weighted moment of initial exponents. A uniform restriction-space estimate and Brunn–Minkowski give a contradictory lower bound. The uniformity is independent of the degree and singularities of the moving center (Section 5).

The resulting point-centered minimizer determines a supporting functional on a finite collection of rational polytopes of normalized section orders. A small ideal perturbation makes discrepancy equality rigid along a reduced divisor S over x . Kawamata–Viehweg vanishing then lifts a nonzero fiber value at x . Only local control near that equality locus is needed; the construction does not assume a globally log canonical boundary (Section 6). Figure 1 records the quantities passed between these steps.

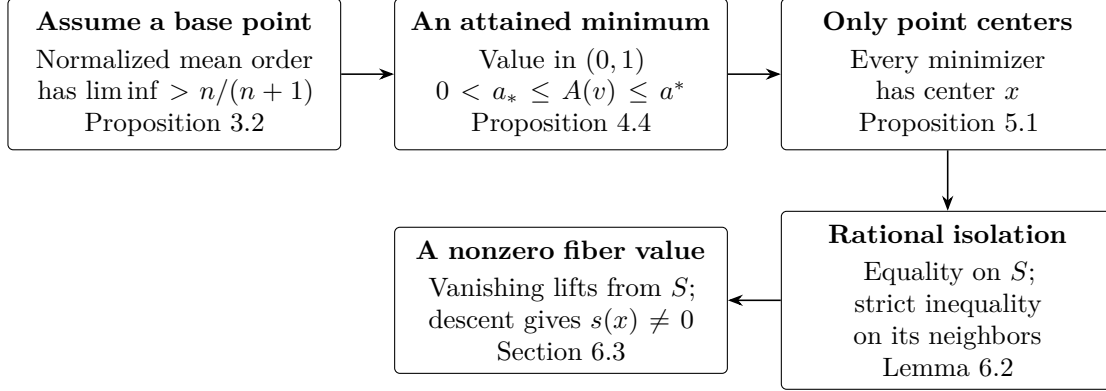


Figure 1: The contradiction proving the sharp adjoint bound. The discrepancy bounds are uniform before fixing $t < n + 1$; for that one parameter, every minimizer has point center in arbitrarily large degrees. Here S is the reduced divisor over x on which the constructed boundary coefficient equals log discrepancy.

2 Orders, thresholds, and vanishing

Throughout the proof, X , L , and n satisfy the hypotheses of Theorem 1.1. Write

$$P = K_X + (n + 1)L, \quad R_j = H^0(X, jL), \quad N_j = \dim R_j.$$

Ampleness gives

$$N_j = \frac{L^n}{n!} j^n + O(j^{n-1}), \quad L^n \geq 1. \quad (2.1)$$

All centers are *closed* centers. Thus a divisor centered along a subvariety W is an admissible test at every point of W , not just at its generic point.

2.1 Weighted orders

We use resolution and principalization in characteristic zero with a prescribed simple normal crossing (SNC) boundary; see [24, Theorems 1.0.1 and 2.4.1]. Models used for vanishing are projective. We may resolve above an already chosen model and retain any previously extracted prime by its strict transform.

For a prime divisor F on a smooth birational model $\pi : Y \rightarrow X$, the log discrepancy is

$$A(F) = 1 + \text{ord}_F(K_{Y/X}).$$

We permit positive multiples of ord_F and scale discrepancy by the same multiple. Orders of sections mean orders of their effective zero divisors; orders of ideals mean the minimum of the orders of local generators. A local frame pulled back from X may be used in these calculations.

We also use weighted monomial orders. Choose a smooth projective birational model and local regular coordinates adapted to the *entire* relative canonical divisor: every component of that divisor meeting the coordinate neighborhood is a coordinate hyperplane. All coordinate divisors used below, together with that support, must form a simple normal crossing divisor. Choose a subset $E_1, \dots, E_r$ of the coordinate divisors and give it positive real weights $u_1, \dots, u_r$; the remaining coordinates have weight zero. The order of a germ is the least weight of a nonzero monomial in their local

equations, with coefficients in the remaining coordinates, and

$$A(v) = \sum_{j=1}^r u_j A(E_j). \quad (2.2)$$

A coordinate divisor absent from the relative canonical divisor has log discrepancy one. We use a chosen irreducible component of the positive-support stratum, whose generic point avoids all zero-weight coordinate divisors; its closed image is the center on X . This adaptation condition is essential for (2.2). The test with all weights zero is excluded.

For positive rational weights this is a divisorial order up to scale. Indeed, clearing denominators and extracting the corresponding primitive ray is a toroidal modification in regular parameters. The relative canonical order in those parameters is the sum of the primitive weights minus one. Adding the pullback of the previous relative canonical divisor gives (2.2). The extraction can be retained on a smooth projective model by resolution. Approximating weights on a fixed support by positive rational weights approximates both discrepancy and the orders of any fixed finite collection of sections. For the latter assertion only finitely many coordinatewise minimal exponents need be considered.

Lemma 2.1 (Local computation of a monomial order). *A monomial order with fixed positive support can be computed in the Taylor ring at any smooth point of its open stratum. Its value is unchanged by replacing each positive-weight coordinate by a unit multiple or by changing the complementary zero-weight coordinates.*

Proof. For a positive threshold c , the corresponding ideal is generated by the monomials in the positive-weight parameters of weight at least c . These generators are preserved, up to units, under the stated changes. In the completed local ring the ideal is extended from a finite-colength monomial ideal in the positive-weight variables. It is therefore primary to their intersection. Localization at the generic point of that stratum followed by contraction does not change membership. The same conclusion in the regular local ring follows from faithful flatness of completion. In particular a coefficient which vanishes at the chosen closed point, but not identically on the stratum, is still a nonzero series in zero-weight variables; it does not raise the old weighted order. $\square$

Lemma 2.2 (SNC retraction). *Let $D_1, \dots, D_N$ be effective divisors on X , and let $Y \rightarrow X$ be a log resolution of their union. Include all exceptional divisors in its SNC divisor, with components E_j . For any positive multiple v of a prime-divisor order over Y , set $u_j = v(E_j)$. Then*

$$v(D_i) = \sum_j u_j \operatorname{ord}_{E_j}(D_i), \quad A(v) \geq \sum_j u_j A(E_j). \quad (2.3)$$

If some u_j is nonzero, the monomial order with these weights at the stratum containing the generic center of v has the same orders on the D_i , no larger discrepancy, and a closed center on X containing that of v .

Proof. Each pulled-back divisor is locally a monomial times a unit, which gives the first equality. The reduced SNC divisor on Y is log canonical: equivalently, a logarithmic volume form has at most a simple pole along every prime on a higher smooth model. Thus the log discrepancy over Y is at least $\sum_j u_j$. Adding the order of $K_{Y/X}$ gives the second inequality and the claimed discrepancy of the monomial order. The stratum closure contains the generic center, so its proper image contains the original closed center. $\square$

2.2 Thresholds in families

For a coherent ideal $\mathfrak{a}$ on a smooth variety, its local log canonical threshold at x is computed on a log resolution by

$$\mathrm{lct}_x(\mathfrak{a}) = \min_{x \in \pi(F), \mathrm{ord}_F(\mathfrak{a}) > 0} \frac{A(F)}{\mathrm{ord}_F(\mathfrak{a})}. \quad (2.4)$$

The minimum is $+\infty$ for the unit ideal. For the zero ideal the threshold is zero. The same formula follows by change of variables from local integrability of $(\sum |g_i|^2)^{-c}$ for generators g_i of $\mathfrak{a}$.

Lemma 2.3 (Closed threshold loci). *Let V be a smooth complex projective variety, let S be a complex algebraic variety, and let $\mathfrak{a}$ be an ideal on $V \times S$ locally given by algebraic families of generators. For a real number c , the locus*

$$\{(x, s) : \mathrm{lct}_x(\mathfrak{a}_s) \leq c\}$$

is Zariski closed. Here $\mathfrak{a}_s$ denotes the ideal generated by the restricted generators, with no flatness requirement on its zero scheme.

Proof. Locally embed the parameter space in a smooth space and extend the finitely many generators holomorphically to a coordinate neighborhood of that ambient space. The analytic semicontinuity of complex singularity exponents applies to $\varphi = \frac{1}{2} \log \sum |g_i|^2$ there and hence on S ; e^φ is locally Hölder continuous. In local coordinates, translation of the variable handles a moving marked point. Consequently the displayed locus is closed in the classical topology; see [5, Lemma 3.2].

It is also algebraically constructible. The following generic-resolution stratification argument parallels the proof of [5, Theorem 3.1]. Resolve the ideal over the generic point of each irreducible component of S and spread the resolution over a dense open subset. After shrinking, generic smoothness makes the fibers and all relevant crossings smooth, so the fiber maps are log resolutions with fixed orders and discrepancy coefficients. Formula (2.4) expresses the threshold condition as a finite union of images of the relevant divisors on this piece. These images are algebraic because the resolution is proper. A finite cover used to separate components preserves constructibility of the image. Induction on the closed complement gives a finite stratification and proves constructibility on $V \times S$. The zero-ideal case is separated as an algebraic condition. In dimension zero the assertion follows directly from the vanishing of the generators.

A constructible subset of a complex algebraic variety which is classically closed is Zariski closed: every Zariski open subset of an irreducible closed subvariety is classically dense there. This proves the assertion. $\square$

2.3 Vanishing and descent

The cohomological input is Kawamata–Viehweg vanishing [18, 23], in the following rounding formulation [7, Theorem 0.1].

Theorem 2.4 (Kawamata–Viehweg vanishing). *Let T be smooth and projective over $\mathbb{C}$. Let Δ be an effective rational SNC divisor with coefficients in $[0, 1)$, and let D be an integral divisor. If $D - (K_T + \Delta)$ is nef and big, then $H^i(T, \mathcal{O}_T(D)) = 0$ for every $i > 0$.*

This is the rounding formulation with $Q = D - K_T - \Delta$: its fractional support is SNC and $K_T + \lceil Q \rceil = D$. In particular, for an integral divisor M and an effective rational SNC divisor B , nefness and bigness of $M - B$ give vanishing for $K_T + M - \lfloor B \rfloor$, by taking $\Delta = \{B\}$.

We will repeatedly use the following elementary descent observation.

Lemma 2.5. *Let $\pi : T \rightarrow V$ be a proper birational morphism of smooth varieties, let P_V be a line bundle on V , and let C be an integral divisor on T whose positive part is exceptional. A section of $\pi^*P_V + C$ defines a regular section of P_V on V . If a prime F on T maps to a point x , the coefficient of C on F is zero, and the section is nonzero generically on F , then the descended section is nonzero at x .*

Proof. View the section as a rational section of P_V on the common function field. Its orders on all nonexceptional primes are nonnegative, since the coefficients of C there are nonpositive. It is therefore regular in codimension one on the normal variety V , and hence everywhere. Under the additional assumptions its pullback has order zero along F . A local function vanishing at x has positive order along every prime centered at x , so the descended section cannot vanish at x . $\square$

3 The ordinary-order alternative

This section converts failure of adjoint generation at a point into a strict asymptotic inequality for ordinary orders. Lattice counting gives the non-strict bound; the main task is to show that its equality case already permits a nonzero fiber value to be lifted.

Fix a closed point $x \in X$. Choose regular parameters at x and a local frame of L , using its powers as frames of jL . We order Taylor monomials first by total degree and then lexicographically, always taking the smallest monomial. Let

$$H_j = \{\text{in}(s) : 0 \neq s \in R_j\} \subset \mathbb{Z}_{\geq 0}^n$$

be the set of initial exponents. Thus $|\text{in}(s)|_1 = \text{mult}_x(s)$.

The equality between the number of initial exponents and the dimension of the section space is the elementary counting principle in [20, Lemma 1.4] and [17, Proposition 2.6]. We include its proof for the order chosen here.

Lemma 3.1. *The set H_j has cardinality N_j , and representatives of its distinct exponents form a basis of R_j . There is a constant $C > 0$, independent of j and s , such that*

$$\text{mult}_x(s) \leq Cj \quad (0 \neq s \in R_j). \quad (3.1)$$

Moreover $H_j + H_q \subset H_{j+q}$.

Proof. Taylor expansion is injective on R_j : a section with zero germ vanishes on a nonempty open set and hence on the integral variety X . Each monomial coefficient has a one-dimensional space of possible values. Successively selecting a smallest initial monomial and eliminating its coefficient from the remaining basis vectors therefore produces a basis with distinct initials. Any linear combination of these vectors has the initial of its first nonzero summand in this ordering. This proves the cardinality assertion.

Choose a fixed sufficiently ample multiple kL . General members of its linear system through x cut out a curve smooth at x and not contained in the divisor of a prescribed nonzero section of jL . Local intersection multiplicity bounds $\text{mult}_x(s)$ from above by the degree of that divisor on the curve, at most $jk^{n-1}L^n$. The same assertion in dimension one is simply the degree bound for an effective divisor on X . This gives (3.1). Finally, the chosen monomial ordering is multiplicative, so initial exponents add under products of sections. $\square$

Proposition 3.2 (Ordinary-order alternative). *If P is not generated by its global sections at x , then*

$$\liminf_{j \rightarrow \infty} \frac{1}{jN_j} \sum_{\gamma \in H_j} |\gamma|_1 > \frac{n}{n+1}. \quad (3.2)$$

Proof. We prove that failure of the strict inequality implies generation at x . The proof has three steps: lattice equality forces volume one, the resulting leading forms give a strict threshold on the exceptional divisor of the point blowup, and vanishing lifts a nonzero section from that divisor. Put $V = L^n$. The number of nonnegative integral points of total degree at most r is $\binom{r+n}{n}$. Filling the points in order of increasing total degree, and using $N_j = Vj^n/n! + O(j^{n-1})$, gives

$$\liminf_{j \rightarrow \infty} \frac{1}{jN_j} \sum_{\gamma \in H_j} |\gamma|_1 \geq V^{1/n} \frac{n}{n+1}. \quad (3.3)$$

Indeed the last occupied degree for a set of N_j smallest points is $V^{1/n}j + O(1)$, and summing total degree over these points gives mean degree $V^{1/n}nj/(n+1) + O(1)$. Since V is a positive integer, failure of (3.2) forces $V = 1$ and equality along a subsequence.

Let

$$\Delta = \{u \in \mathbb{R}_{\geq 0}^n : |u|_1 \leq 1\}.$$

Along this subsequence, every fixed open ball with closure in the interior of Δ meets $j^{-1}H_j$ for all sufficiently large j . To see this, compare H_j with a set T_j of N_j smallest lattice points. If such a ball were missed, at least c_1j^n points of T_j of degrees at most $(1-c_2)j$ would be absent, for fixed positive c_1, c_2 . Every replacement outside T_j has degree at least $j - O(1)$. The remaining replacements cannot decrease the degree sum, by the definition of T_j . The normalized mean would therefore exceed the minimum by a fixed positive amount, contradicting equality along the subsequence.

Choose a rational number t' with $n < t' < n+1$ and put $c = (1/t', \dots, 1/t')$. This point is in the interior of Δ . Choose finitely many small balls whose closures lie there and such that any choice of one point from each ball has convex hull containing c in its interior. For one sufficiently large degree j on the subsequence, choose initial exponents $\gamma_\nu \in H_j$ in these balls after scaling by j^{-1} . The rational polytope

$$Q = \text{conv}\{\gamma_\nu/j\}$$

contains c in its interior. Its intersection with $\{|u|_1 = n/t'\}$ has rational points $p_1, \dots, p_b$ whose convex hull contains c in its interior relative to this full slice hyperplane. In dimension one this slice is the singleton $\{c\}$.

Write each p_ℓ as a rational convex combination of the γ_ν/j , and choose sections s_ν with those initials. If $t' = a/b'$ in lowest terms, choose a positive integer Q_0 divisible by a and by every denominator in these convex combinations. Taking the corresponding products of Q_0 sections gives sections $r_1, \dots, r_b$ of jQ_0L . Set $h = jQ_0/t'$, which is a positive integer by construction. Their initial exponents β_ℓ satisfy

$$r_\ell \in H^0(X, ht'L), \quad \beta_\ell = jQ_0p_\ell, \quad |\beta_\ell|_1 = nh. \quad (3.4)$$

In particular, $\text{conv}\{\beta_\ell/h\}$ contains $\mathbf{1} = (1, \dots, 1)$ in relative interior in the full hyperplane $\{|u|_1 = n\}$.

Let $p : B = \text{Bl}_x X \rightarrow X$ and $E \simeq \mathbb{P}^{n-1}$ be the exceptional divisor. When $n = 1$, take p to be the identity and $E = x$. Removing the common order nh from the pullbacks of the r_ℓ gives a subsystem of

$$H = p^*(ht'L) - nhE$$

with base ideal $\mathfrak{b}$. Its restriction to E is generated, in local frames, by the leading homogeneous forms g_ℓ of the r_ℓ , each of degree nh . We claim that

$$\text{lct}_z(E, \mathfrak{b}|_E) > 1/h \quad (z \in E). \quad (3.5)$$

Here the restricted ideal means its image in $\mathcal{O}_E$. For $n = 1$ it is the unit ideal, so the assertion holds immediately.

The monomial threshold step below is the unit-monomial special case of Howald's multiplier-ideal formula [16, Main Theorem]; we give the direct SNC argument in this setting.

For $n > 1$, first replace the g_ℓ by their initial monomials z^{β_ℓ} . On the affine chart of E where $z_k \neq 0$, their exponents are obtained by omitting coordinate k . Projection from the hyperplane $|u|_1 = n$ is an affine isomorphism, so the resulting normalized exponent polytope contains the all-ones vector in ordinary interior. Consequently, for every nonzero $u \in \mathbb{R}_{\geq 0}^{n-1}$,

$$\frac{1}{h} \min_\ell \langle u, \beta_\ell^{(k)} \rangle < \sum_i u_i. \quad (3.6)$$

For a prime-divisor order over this chart, use its values on the coordinate functions as u . The left side is the order of the monomial ideal divided by h , while the right side is at most the log discrepancy, by the SNC inequality in Lemma 2.2. If $u = 0$, the ideal order is zero and the discrepancy is positive. Thus the monomial ideal is klt at coefficient $1/h$. Computing its threshold on a log resolution, where a finite minimum suffices, gives a strict threshold greater than $1/h$ at every point.

This statement transfers to the original forms by an algebraic degeneration. Since only finitely many monomials occur, choose an integral weight vector w which makes the lexicographic initial of each g_ℓ its unique term of smallest weight. The forms

$$g_{\ell,\lambda}(z) = \lambda^{-\langle w, \beta_\ell \rangle} g_\ell(\lambda^{w_1} z_1, \dots, \lambda^{w_n} z_n)$$

have polynomial coefficients in λ , with monomial special fiber. They define an algebraic family of ideals on $E \times \mathbb{A}^1$. By Lemma 2.3, the locus where the fiber threshold is at most $1/h$ is closed. Its image in $\mathbb{A}^1$ is closed because E is proper, and does not contain 0. For some nonzero parameter the threshold is therefore greater than $1/h$ everywhere on E . All nonzero fibers are obtained from the original forms by a diagonal automorphism of E and nonzero scalar changes of generators. This proves (3.5). This application uses precisely the family semicontinuity of complex singularity exponents [5] incorporated in Lemma 2.3.

It remains to lift from E . The blowup formula gives

$$p^*P - (K_B + E) - H/h = (n + 1 - t')p^*L. \quad (3.7)$$

Take a projective log resolution $\sigma : T \rightarrow B$ of E and $\mathfrak{b}$, with strict transform E' and $\mathfrak{b}\mathcal{O}_T = \mathcal{O}_T(-G)$. The divisor E' is not a component of G , since the restricted system is not identically zero. Set

$$D_T = K_T + \sigma^*(p^*P - K_B - E) - \lfloor G/h \rfloor.$$

This is an integral divisor, and

$$D_T - K_T - \{G/h\} = (n + 1 - t')(p\sigma)^*L + (\sigma^*H - G)/h. \quad (3.8)$$

The first summand is nef and big and the second is nef, because $\sigma^*H - G$ is globally generated. The fractional boundary has SNC support. Theorem 2.4, in the stated logarithmic form of Kawamata–Viehweg vanishing [7], therefore gives $H^1(T, \mathcal{O}_T(D_T)) = 0$.

Let $\sigma_E : E' \rightarrow E$ be the induced morphism. Adjunction identifies the restriction of $D_T + E'$ with

$$\sigma_E^*(p^*P|_E) + K_{E'/E} - \lfloor G|_{E'}/h \rfloor. \quad (3.9)$$

Indeed $E' + \text{Supp}(G)$ is SNC. Each component restricts with multiplicity one, and two distinct components cannot restrict to the same prime divisor on E' , so taking floors commutes with restriction. The generators on E' generate $\mathcal{O}_{E'}(-G|_{E'})$, the pullback of the restricted ideal. The strict threshold in (3.5) implies

$$K_{E'/E} - \lfloor G|_{E'}/h \rfloor \geq 0.$$

The first line bundle in (3.9) is trivial with fiber P_x . A nonzero vector in this fiber, multiplied by the canonical section of the effective correction, gives a section on E' nonzero at its generic point. The exact sequence for E' and the vanishing just proved lift it to a section of $D_T + E'$.

Relative to σ^*p^*P , the correction of this lifted bundle is

$$C_T = K_{T/B} - \sigma^*E + E' - \lfloor G/h \rfloor.$$

Its coefficient on E' is zero, and its coefficients on every other σ -nonexceptional divisor are nonpositive. Its positive part is therefore exceptional over X . Since E' maps to x and the lifted section is nonzero at its generic point, Lemma 2.5, applied to $p\sigma$, gives a section of P nonzero at x . This proves the proposition, including dimension one, when E' is the point itself. $\square$

4 Finite-degree exponential minimization

We now allow both the basis and the monomial test to vary. The strict ordinary-order inequality makes the objective less than one, and a supply of low-order jets bounds the discrepancy on that sublevel set. To prove attainment, we first select one flag and its adapted basis; only then do we choose the resolution on which compactness is used.

For $N_m > 0$, an ordered basis $\mathbf{s} = (s_1, \dots, s_{N_m})$ of R_m , a nonzero monomial test v , and $t > 0$, put

$$F_{m,t}(v, \mathbf{s}) = \frac{1}{N_m} \sum_{i=1}^{N_m} \exp\left(\frac{A(v)}{t} - \frac{v(s_i)}{m}\right), \quad f_{m,t}(z) = \inf_{\substack{v, \mathbf{s} \\ z \in c_X(v)}} F_{m,t}(v, \mathbf{s}). \quad (4.1)$$

Here $c_X(v)$ denotes the closed center, and the infimum also ranges over the bases. Positive rescaling of a test is always allowed. Approximating its positive weights by rational weights preserves its center and approximates the orders of finitely many sections and its discrepancy. Thus the same infimum is obtained using positive multiples of prime-divisor orders.

Lemma 4.1. *If the center of a test v contains x , then every nonzero section s satisfies*

$$v(s) \leq A(v) \text{mult}_x(s). \quad (4.2)$$

In particular, for $s \in R_m$ one has $0 \leq v(s)/A(v) \leq Cm$, with C as in Lemma 3.1.

Proof. For an effective divisor with local equation g of multiplicity $r > 0$ at a smooth point, the elementary bound $\text{lct}_x(g) \geq 1/r$ suffices. One analytic proof uses Weierstrass preparation in a general coordinate direction. On a smaller polydisc, g is a unit times a monic polynomial of degree r in that coordinate. For any $c < 1/r$, factor this polynomial into its r complex roots for each fixed choice of the other coordinates. Hölder's inequality bounds the integral of its absolute value to the power $-2c$ by the product of integrals with exponents $-2cr$. These one-variable integrals are uniformly finite because $cr < 1$ and the roots remain bounded. Fubini's theorem proves local integrability. The unit case has infinite threshold. Computing the threshold on a log resolution now gives (4.2) for divisorial orders with centers containing x , and rational approximation gives it for the admitted monomial tests. $\square$

Lemma 4.2 (Uniform sublevel bounds). *Suppose that P is not generated at x . There are numbers $0 < t_0 < n + 1$, $c > 0$, $0 < \delta < 1$, $0 < a_* \leq a^* < \infty$, and an integer m_0 such that, for every $m \geq m_0$ and every $t \in [t_0, n + 1]$,*

$$c \leq f_{m,t}(x) \leq 1 - \delta. \quad (4.3)$$

Moreover, any pair in (4.1) centered with image containing x and satisfying $F_{m,t}(v, \mathbf{s}) \leq 1 - \delta/2$ has

$$a_* \leq A(v) \leq a^*. \quad (4.4)$$

All these constants are independent of m and t in the indicated ranges.

Proof. By Proposition 3.2, there is $\epsilon > 0$ such that, for all sufficiently large m ,

$$\frac{1}{mN_m} \sum_{\gamma \in H_m} |\gamma|_1 \geq \frac{n}{n+1} + 3\epsilon.$$

Choose $t_0 < n + 1$ sufficiently close that $n/t_0 \leq n/(n+1) + \epsilon$. Take a basis with distinct ordinary initials, and use $v = \lambda \text{ord}_x$, where ord_x is the ordinary point order and $A(\text{ord}_x) = n$. In dimension one this is the order of the point itself. The linear coefficient at $\lambda = 0$ in its objective is at most -2ϵ , uniformly in m and t . By (3.1), the quantities $n/t - \text{mult}_x(s_i)/m$ are uniformly bounded, so the quadratic remainder in their exponential expansion is uniformly $O(\lambda^2)$. A fixed sufficiently small $\lambda > 0$ therefore gives $F_{m,t} \leq 1 - \delta$ for a fixed $\delta > 0$.

We next give a uniform supply of low-order jets. Choose $k > 0$ such that kL is very ample, and sections $s_0, s_1, \dots, s_n$ of kL such that $s_0(x) \neq 0$ and $z_i = s_i/s_0$ are regular parameters at x . For each residue class modulo k , choose a fixed sufficiently large integer b_r in that class for which $b_r L$ is globally generated, and choose $\tau_r \in H^0(X, b_r L)$ nonzero at x . Write $m = qk + b_r$ using the corresponding residue. For $|\alpha|_1 \leq q$, the sections

$$\tau_r s_0^{q-|\alpha|_1} s_1^{\alpha_1} \dots s_n^{\alpha_n}$$

have germs equal to a common unit times z^α in a local frame of mL . In particular, if

$$r_m = \min \left(q, \left\lfloor \frac{m}{2(n+1)} \right\rfloor \right),$$

their jets for $|\alpha|_1 \leq r_m$ are linearly independent modulo ordinary order greater than $m/(2t)$. For large m , r_m is at least a fixed positive multiple of m . The Hilbert asymptotic consequently gives a constant $0 < \kappa < 1$, independent of m, t , with

$$\text{codim}_{R_m} \{s : \text{mult}_x(s) > m/(2t)\} \geq \kappa N_m. \quad (4.5)$$

Put $a = A(v)$. By Lemma 4.1, the subspace $\{s : v(s) > ma/(2t)\}$ is contained in the subspace in (4.5). Thus at least κN_m vectors of any basis, whether or not it is adapted to v , have $v(s_i) \leq ma/(2t)$. It follows that

$$F_{m,t}(v, \mathbf{s}) \geq \kappa \exp\left(\frac{a}{2t}\right) \geq \kappa \exp\left(\frac{a}{2(n+1)}\right). \quad (4.6)$$

This proves the lower bound in (4.3), with $c = \kappa$, and bounds a above on the stated sublevel set.

Finally, (4.2) and (3.1) show that $F_{m,t}(v, \mathbf{s}) \geq e^{-Ca}$. Choose $a_* > 0$ sufficiently small that $e^{-Ca_*} > 1 - \delta/2$. No pair in the sublevel set can have $a < a_*$. Enlarging the upper bound if necessary gives $a^* \geq a_*$ and proves all the assertions. $\square$

Lemma 4.3 (Closed realizability loci). *Fix m with $N_m > 0$ and a decreasing list $\alpha_1 \geq \dots \geq \alpha_{N_m} \geq 0$ of rational numbers. The set of points $z \in X$ for which some ordered basis and some prime-divisor order v , with $z \in c_X(v)$, satisfy*

$$\frac{v(s_i)}{A(v)} \geq \alpha_i \quad (1 \leq i \leq N_m) \quad (4.7)$$

is Zariski closed. There is also a closed incidence locus in the product of X with the complete flag variety of R_m recording realizability by bases adapted to a given flag.

Proof. Let $\mathcal{F}$ be that projective flag variety. A flag has subspaces $U_i \subset R_m$ of dimensions i and associated base ideals $\mathfrak{b}_i \subset \mathcal{O}_X$, obtained by evaluation after twisting by $-mL$. If (4.7) holds, the flag spanned by the first i basis vectors satisfies

$$v(\mathfrak{b}_i) \geq \alpha_i A(v) \quad (1 \leq i \leq N_m), \quad (4.8)$$

since the list is decreasing. Conversely these inequalities imply (4.7) for every basis adapted to the flag.

If at least one α_i is positive, choose a positive integer M such that M/α_i is integral for every positive entry, and set

$$\mathfrak{a}_\alpha = \sum_{i:\alpha_i>0} \mathfrak{b}_i^{M/\alpha_i}.$$

The simultaneous inequalities (4.8) are equivalent to $v(\mathfrak{a}_\alpha) \geq MA(v)$. A divisorial witness with center containing z exists if and only if

$$\text{lct}_z(X, \mathfrak{a}_\alpha) \leq 1/M, \quad (4.9)$$

because the local threshold is computed by a component on a log resolution. The universal subbundles on $\mathcal{F}$ give algebraic families of generators for the base ideals on $X \times \mathcal{F}$, and hence for their indicated powers and sum. Lemma 2.3 applied to (4.9) gives the closed incidence locus. Its projection to X is closed by properness of $\mathcal{F}$. If all entries vanish, both loci are the whole parameter spaces. $\square$

Proposition 4.4 (Existence and uniform bounds for minimizers). *Suppose that P is not generated at x . For the constants and ranges in Lemma 4.2, $f_{m,t}(x)$ satisfies (4.3) and is attained by an ordered basis of R_m and a nonzero real-weight monomial test on a projective log resolution of the divisors of that basis. Every minimizing test has*

$$a_* \leq A(v) \leq a^*.$$

More generally the same bounds hold eventually along every minimizing sequence. In particular, the positive lower bound a_ is independent of $m \geq m_0$ and $t \in [t_0, n+1]$.*

Proof. Only attainment remains to be proved. Fix m, t in the indicated ranges and take a sequence of divisorial pairs whose objectives decrease to $f_{m,t}(x)$. Eventually they lie in the sublevel set of Lemma 4.2. Order each basis by decreasing normalized values and pass to a subsequence for which

$$a_b = A(v_b) \longrightarrow a \in [a_*, a^*], \quad q_{b,i} = \frac{v_b(s_{b,i})}{A(v_b)} \longrightarrow q_i.$$

This is possible because $0 \leq q_{b,i} \leq Cm$. The limit list is decreasing and

$$f_{m,t}(x) = \frac{1}{N_m} \sum_i \exp\left(\frac{a}{t} - \frac{aq_i}{m}\right). \quad (4.10)$$

At least one q_i is positive: otherwise the right side would be $e^{a/t} > 1$, contrary to (4.3).

Choose decreasing nonnegative rational lists $\alpha^{(k)} = (\alpha_i^{(k)})$ which increase componentwise to (q_i) and are strictly below q_i whenever $q_i > 0$. Each list is satisfied by all sufficiently late pairs in the minimizing sequence. By the incidence assertion of Lemma 4.3, the flags admitting a divisorial witness at x for that list form a nonempty closed subset $\mathcal{F}_k$ of $\mathcal{F}$. The subsets are nested, so projectivity gives a flag in their intersection. Fix a basis $s_1, \dots, s_{N_m}$ adapted to this one flag.

Resolve the divisors of these sections on a single smooth projective model $Y \rightarrow X$, including the exceptional divisors in the SNC support. For each sufficiently large k , choose a divisorial witness for the fixed flag and the list $\alpha^{(k)}$, and exhibit it on a common smooth model dominating Y . Its retraction to this fixed SNC model, as in Lemma 2.2, preserves all the section orders, cannot increase discrepancy, and has center on X containing x . The retraction is nonzero because the list has a positive entry. Normalize it to discrepancy one. It then satisfies

$$\tilde{v}_k(s_i) \geq \alpha_i^{(k)} \quad (1 \leq i \leq N_m). \quad (4.11)$$

There are finitely many SNC strata and their components on Y . After passing to a subsequence, the retractions use the same stratum component, with coordinate divisors indexed by a fixed set J . Their normalized weights belong to the compact simplex

$$\left\{ (u_j)_{j \in J} : u_j \geq 0, \quad \sum_{j \in J} A(E_j) u_j = 1 \right\};$$

all coefficients $A(E_j)$ are strictly positive because X is smooth. Pass to a limit of the weights. A weight may become zero. We take the component of the remaining positive-support intersection that contains the old stratum component. Its closure contains the old center, so its image still contains x . The limit is a nonzero monomial test $\tilde{v}$ of discrepancy one. On the fixed resolution, the section orders are linear in these weights. Consequently (4.11) gives $\tilde{v}(s_i) \geq q_i$ for every i .

Set $v = a\tilde{v}$. Comparing with (4.10) yields

$$F_{m,t}(v, \mathbf{s}) \leq \frac{1}{N_m} \sum_i \exp\left(\frac{a}{t} - \frac{aq_i}{m}\right) = f_{m,t}(x).$$

The reverse inequality holds by definition of the infimum, so this pair attains it. The claimed discrepancy bounds for all minimizers and eventually for minimizing sequences follow directly from the uniform sublevel bounds. $\square$

The same closed realizability loci also compare the infimum along a possible center. This comparison will allow the tangential variation in Section 5 to be tested at a very general point.

Lemma 4.5 (Specialization comparison). *For fixed m with $N_m > 0$, let $W \subset X$ be a closed irreducible subvariety containing x . At a very general closed point $y \in W$, one has, for every $t > 0$,*

$$f_{m,t}(y) \geq f_{m,t}(x). \quad (4.12)$$

Proof. There are countably many rational lists of the kind in Lemma 4.3. For each, intersect its closed realizability locus with W , and exclude that intersection if it is proper in W . The complement contains very general closed points over $\mathbb{C}$. For any remaining y , every such list realized at y is realized at x .

Take a divisorial test and a basis at y , reorder the basis so that $q_i = v(s_i)/A(v)$ is decreasing, and put $a = A(v) > 0$. Approximate (q_i) componentwise from below by decreasing rational lists

$(\alpha_i^{(b)})$. For each list, choose a realizing pair at x and scale its test to discrepancy exactly a . Its objective is at most

$$\frac{1}{N_m} \sum_i \exp\left(\frac{a}{t} - \frac{a\alpha_i^{(b)}}{m}\right).$$

Taking the limit gives $f_{m,t}(x) \leq F_{m,t}(v, \mathbf{s})$. Infimizing over the divisorial pairs at y proves (4.12). The excluded loci do not depend on t , which explains the simultaneous statement for all positive t . $\square$

5 Eliminating positive-dimensional centers

The specialization comparison can be combined with a tangential variation of a minimizing test. The essential point is that the estimates for sections on its center are uniform as that center varies.

Proposition 5.1. *Suppose that P is not generated at x . There is a real number $0 < t < n + 1$ such that, for arbitrarily large integers m , the infimum $f_{m,t}(x)$ is attained and every minimizing monomial pair has center exactly $\{x\}$.*

Here a minimizing pair consists of a monomial test and a basis attaining the infimum. The assertion concerns all such pairs, even if their models, coordinates, and bases differ. This universal assertion will be needed in Section 6: a second test with the same normalized section orders and discrepancy is also minimizing, so it too must have point center. We first establish a uniform estimate that will apply to all possible centers.

5.1 A uniform estimate on restriction spaces

Fix a positive integer k such that kL is very ample, and use its complete system to embed X in a projective space. For an integral d -dimensional closed subvariety $W \subset X$, put

$$I_W(h) = \dim_{\mathbb{C}} \operatorname{im}(R_h \longrightarrow H^0(W, hL|_W)).$$

We require a lower estimate for this image, rather than surjectivity of the restriction map.

Lemma 5.2. *For each $1 \leq d \leq n$, there are constants C_d and H_d , depending only on (X, L, k, d) , such that*

$$(d! I_W(h))^{1/d} \geq h - C_d \quad (h \geq H_d) \tag{5.1}$$

for every integral d -dimensional closed subvariety $W \subset X$.

Proof. Set $D_0 = k^d$. The degree of W in the fixed embedding is

$$D = (kL)^d \cdot W = k^d(L^d \cdot W) \geq D_0,$$

because the last intersection number is a positive integer. A general linear projection gives a finite morphism $\pi : W \longrightarrow \mathbb{P}^d$ of degree D . Its generic fiber is separable, since the ground field has characteristic zero. Write $K = \mathbb{C}(\mathbb{P}^d)$ and $E = \mathbb{C}(W)$.

Choose a nonzero linear coordinate s_0 of the projection. All points of the geometric generic fiber lie in $s_0 \neq 0$. Select D_0 of these distinct points. Over an algebraic closure of K , affine linear forms separate any two of them. For each selected point, multiplying at most $D_0 - 1$ such forms gives a polynomial vanishing at the other selected points and not at that one. Homogenizing with s_0 gives ambient forms of degree D_0 whose normalized values span the functions on these D_0 points.

Let V_{D_0} be the complex vector space of ambient degree- D_0 forms. Restriction and division by $s_0^{D_0}$ define a K -linear map $V_{D_0} \otimes_{\mathbb{C}} K \rightarrow E$. It has rank at least D_0 . Indeed this rank can be checked after extending scalars to the algebraic closure of K , where the preceding evaluation argument applies. We may therefore choose ambient degree- D_0 forms $F_1, \dots, F_{D_0}$ for which

$$F_1/s_0^{D_0}, \dots, F_{D_0}/s_0^{D_0} \in E$$

are linearly independent over K . The forms can be chosen among a fixed monomial spanning set; their coefficients need not depend on elements of K .

For $q \geq D_0$, multiply each F_i by a monomial basis of the homogeneous forms of degree $q - D_0$ in the coordinates of the projection. These products restrict to linearly independent sections on W . In fact, a relation between them, divided by s_0^q , is a relation between the displayed K -independent elements; all its coefficients must vanish in K . The corresponding base forms must then vanish identically on $\mathbb{P}^d$. The products are restrictions of sections in R_{qk} , so

$$I_W(qk) \geq D_0 \binom{q - D_0 + d}{d}. \quad (5.2)$$

In particular,

$$(d! I_W(qk))^{1/d} \geq k(q - D_0) = qk - k^{d+1}. \quad (5.3)$$

To cover all degrees, choose, once and for all, an integer h_r in each residue class r modulo k such that $h_r L$ is globally generated. There is a section of $h_r L$ not identically zero on any given W . Multiplication by that section injects the degree- qk restriction space into the degree- $(qk + h_r)$ restriction space, because W is integral. For $h = qk + h_r$ and $q \geq D_0$, Equation (5.3) thus gives

$$(d! I_W(h))^{1/d} \geq h - (h_r + k^{d+1}).$$

Taking maxima over the finitely many residue classes proves the lemma. For example, both C_d and H_d may be taken to equal $\max_r h_r + k^{d+1}$. All thresholds and constants are independent of W . $\square$

5.2 A minimizing test and its tangent directions

Assume from now on that P is not generated at x . Fix $t \in [t_0, n + 1]$ and a sufficiently large degree m in the ranges of Proposition 4.4, and write $f_j = f_{j,t}(x)$. Suppose that a minimizing monomial pair has test v and center W of dimension $d > 0$. We derive upper and lower bounds for its weighted tangential moment, keeping m and t fixed in this calculation. Let Z be the closure of its defining stratum on a smooth model $Y \rightarrow X$, and let r be the number of its positive weights. Put $k' = n - d$, so $r \leq k'$. For the local estimates below, keep this test, its model, and the coordinates chosen below fixed; section spaces in every degree j use the same Taylor order.

Choose a very general smooth point $y \in W$ satisfying Lemma 4.5 in degree m :

$$f_{m,t}(y) \geq f_{m,t}(x). \quad (5.4)$$

The very general condition and the smooth open sets used below can be met simultaneously over $\mathbb{C}$. Above y choose a point z of the open stratum of Z where $Z \rightarrow W$ is smooth, avoiding all additional components of the relative canonical divisor. Generic smoothness allows these choices. Take regular parameters at z in three groups:

1. the r parameters defining the positive-weight crossing components;
2. $k' - r$ parameters along the fiber of $Z \rightarrow W$;

3. d parameters pulled back from affine linear projective coordinates on W vanishing at y . For the last group, choose d independent differentials from the fixed kL embedding. Complete them by the fiber parameters using the smoothness of $Z \rightarrow W$.

The point z lies in the open positive-support stratum, the positive equations change only by units, and the complementary parameters have weight zero. By Lemma 2.1, the original weighted order is therefore unchanged and can be computed from Taylor series at z .

Choose a local frame of L near y and pull it back to Y . The terms of a section with all first r exponents zero are its restriction to Z . Since that restriction comes from W , its Taylor expansion depends only on the last d parameters.

Write Taylor exponents as

$$(p, \beta) \in \mathbb{Z}_{\geq 0}^{k'} \times \mathbb{Z}_{\geq 0}^d,$$

and let $\ell(p)$ be their old weighted order; the middle parameters have weight zero. Order monomials first by increasing $\ell(p)$, then by increasing $|\beta|_1$, and finally by a fixed additive monomial well-order, such as total degree followed by lexicographic order. This combined order is a well-order: below any fixed weighted bound there are only finitely many exponent choices in the positive-weight variables; the tangential total degree and the last tie-break then have least elements. It is also additive.

Let Γ_j be the set of possible initial exponents of nonzero sections of R_j in these coordinates. The leading coefficient has one-dimensional leaves, so elimination gives

$$\#\Gamma_j = N_j, \quad \Gamma_j + \Gamma_q \subset \Gamma_{j+q}. \quad (5.5)$$

The cardinality follows by the elimination argument in the proof of Lemma 3.1; compare the one-dimensional-leaf count in [17, Proposition 2.6]. Multiplication adds initials because the order is additive and the local frames in all degrees are powers of the same frame.

Take a degree- m basis representing the distinct elements of Γ_m . Since the initial order first compares v -values, this basis is adapted to the v -filtration: its sorted valuation values dominate those of any basis. Replacing the original minimizing basis by this one cannot increase the objective, so the new pair still realizes the global minimum. For this fixed test and reference degree m , put

$$Q_j = \sum_{(p, \beta) \in \Gamma_j} e^{-\ell(p)/m}. \quad (5.6)$$

In particular,

$$Q_m = N_m f_m e^{-A(v)/t}. \quad (5.7)$$

We first use minimality to bound the weighted mean tangential degree in Γ_m from above.

Perturb v at z by giving weight ϵ to each of the d tangential parameters and weight ϵ^2 to each of the $k' - r$ fiber parameters, while leaving its positive weights unchanged. For $\epsilon > 0$ this is an allowed monomial test v_ϵ with center z on the model and center y on X . Since the added coordinate divisors have discrepancy one near z ,

$$A(v_\epsilon) = A(v) + d\epsilon + (k' - r)\epsilon^2. \quad (5.8)$$

For a section of the chosen basis with initial exponent (p, β) ,

$$v_\epsilon(s) = \ell(p) + \epsilon|\beta|_1 + O(\epsilon^2) \quad (\epsilon \downarrow 0). \quad (5.9)$$

To justify this for Taylor series, fix the section. There are only finitely many positive-coordinate exponent choices below any fixed old weighted bound, so there is a positive gap from its least old weight to the next larger weight in a bounded interval. Nonnegativity of every exponent then

excludes higher old weights from the minimum for sufficiently small ϵ . Among terms of least old weight, the least tangential total degree determines the coefficient of ϵ ; its value is $|\beta|_1$. Remaining differences affect only the ϵ^2 coefficient. One term realizing both least quantities provides a finite upper bound for that coefficient. The claim follows. There is no need to choose ϵ uniformly as m varies: the differentiation is performed for one fixed finite basis at a time.

Let $F(\epsilon)$ be the objective of that basis and v_ϵ in degree m . By (5.4),

$$F(\epsilon) \geq f_{m,t}(y) \geq f_{m,t}(x) = F(0).$$

Its right derivative at zero is therefore nonnegative. Differentiating using (5.8) and (5.9), and canceling positive common factors, gives

$$T_m := \frac{\sum_{(p,\beta) \in \Gamma_m} |\beta|_1 e^{-\ell(p)/m}}{mQ_m} \leq \frac{d}{t}. \quad (5.10)$$

5.3 Growth forced by restricted sections

We now bound the same moment T_m from below. Restrictions to W give many pure tangential initials; multiplication by their sections forces every nonempty transverse slice to grow with the degree. For a fixed transverse exponent p , define

$$G_p(j) = \{\beta : (p, \beta) \in \Gamma_j\}, \quad b_p(j) = (d! \#G_p(j))^{1/d},$$

with $b_p(j) = 0$ for an empty slice. The restriction observation above implies

$$\#G_0(h) \geq I_W(h).$$

Indeed a nonzero restriction has old weight zero and no fiber variables; elimination within the restriction space gives distinct pure tangential initials. Lemma 5.2 consequently gives

$$b_0(h) \geq h - C_d \quad (5.11)$$

for all sufficiently large h , uniformly in the center and the test.

There is also a useful exact inclusion in bounded degree. Set $h_0 = kd$. Write the selected affine projective coordinates as s_i/s_0 , with $s_0(y) \neq 0$ and $s_i(y) = 0$, for $1 \leq i \leq d$. For $\beta \in \{0, 1\}^d$, the section

$$\prod_{i=1}^d s_i^{\beta_i} s_0^{d-|\beta|_1} \in R_{h_0}$$

has initial exponent $(0, \beta)$. The change from the projective trivialization to our local frame is a unit, whose initial exponent is zero. Hence

$$\{0, 1\}^d \subset G_0(h_0). \quad (5.12)$$

The cube inclusion turns the continuous Brunn–Minkowski inequality into the discrete estimate needed here. If A, C are nonempty finite subsets of $\mathbb{Z}^d$, then

$$(A + [0, 1]^d) + (C + [0, 1]^d) \subset A + C + \{0, 1\}^d + [0, 1]^d.$$

The volume of a finite lattice set thickened by the unit cube is its cardinality, since different cubes have disjoint interiors. The Brunn–Minkowski inequality [10, Theorem 4.1], applied to these finite unions of cubes, gives

$$\#(A + C + \{0, 1\}^d)^{1/d} \geq (\#A)^{1/d} + (\#C)^{1/d}. \quad (5.13)$$

Apply this with $A = G_p(j)$ and $C = G_0(h - h_0)$. Equations (5.5) and (5.12) show that their sum with the cube lies in $G_p(j + h)$. Thus, whenever $G_p(j)$ is nonempty and h is sufficiently large,

$$b_p(j + h) \geq b_p(j) + h - C'_d, \quad C'_d = C_d + h_0. \quad (5.14)$$

The constants and degree threshold remain independent of j , p , the center, and its chosen coordinates.

To sum these slice-growth estimates with the exponential weights, put

$$B_j = \frac{1}{d!} \sum_p e^{-\ell(p)/m} b_p(j)^{d+1},$$

using the same fixed test and reference degree m as in (5.6). All these sums are finite. For sufficiently large h , we have $h - C'_d \geq 0$. Convexity of the $(d + 1)$ st power and (5.14) yield

$$B_{j+h} \geq B_j + (d + 1)(h - C'_d)Q_j. \quad (5.15)$$

Only slices present at degree j are needed in deriving this estimate; new slices make a nonnegative contribution. The coefficient Q_j appears because $b_p(j)^d/d! = \#G_p(j)$.

The power $d + 1$ in B_j relates slice size to tangential moment. For any finite $A \subset \mathbb{Z}_{\geq 0}^d$, let $b = (d! \#A)^{1/d}$. Among measurable subsets of the positive orthant of volume $\#A$, the simplex

$$\{u_i \geq 0 : u_1 + \cdots + u_d \leq b\}$$

minimizes the integral of $u_1 + \cdots + u_d$. This follows directly by comparing the integrand inside and outside this sublevel set. Its integral is $db^{d+1}/((d + 1)d!)$. On the other hand, the integral over $A + [0, 1]^d$ is $\sum_{\beta \in A} (|\beta|_1 + d/2)$. Enlarging $d/2$ to d , applying the resulting inequality slice by slice, and inserting the weights gives

$$\sum_{(p, \beta) \in \Gamma_m} e^{-\ell(p)/m} (|\beta|_1 + d) \geq \frac{d}{d + 1} B_m. \quad (5.16)$$

5.4 The limiting contradiction

For each fixed center dimension $d > 0$, the preceding bounds hold for every minimizing pair in the indicated ranges, with slice-growth constants independent of its center and degree. We now complete the proof of Proposition 5.1. By Propositions 3.2 and 4.4, there is a common interval of values of t immediately below $n + 1$ on which the minima, for sufficiently large degrees, are bounded away from zero and one, and their discrepancies lie in a fixed compact interval $[a_*, a^*] \subset (0, \infty)$. In particular,

$$I_* := \int_0^1 s^n \exp\left(\frac{(1-s)a_*}{n+1}\right) ds > \frac{1}{n+1}.$$

Choose once and for all $t < n + 1$ in that interval sufficiently close to $n + 1$ that

$$I_* > \frac{1}{t}. \quad (5.17)$$

Keep this choice of t in $f_j = f_{j,t}(x)$. Put $c = \liminf_{j \rightarrow \infty} f_j > 0$, and choose an increasing sequence of degrees m for which $f_m \rightarrow c$.

Suppose, seeking a contradiction, that infinitely many of these degrees admit a minimizing monomial pair with positive-dimensional center. Choose one such pair in each degree and pass to a

subsequence on which the center dimension is a fixed $d > 0$. The centers, valuations, and models are allowed to vary along this subsequence.

Apply the preceding construction to each chosen pair. Its test and coordinates may change with m , while all auxiliary degrees j for that m use that one test and coordinate system.

Fix an integer $l \geq 2$. For $1 \leq i \leq l$, put $j_i = \lfloor mi/l \rfloor$, so $j_l = m$. Table 1 summarizes the dependencies and order of choices established above.

As m tends to infinity on our chosen subsequence, all increments $j_{i+1} - j_i$ tend to infinity. Thus (5.15) applies successively. Starting with $B_{j_1} \geq 0$ and then using (5.16) yields

$$T_m + \frac{d}{m} \geq d \sum_{i=1}^{l-1} \frac{j_{i+1} - j_i - C'_d}{m} \frac{Q_{j_i}}{Q_m}. \quad (5.18)$$

The uniformity of C'_d is indispensable here, since the center and the model may depend on m .

For the auxiliary degrees, the test $(j/m)v$ is admissible at x , because its center is still W . Its discrepancy is $jA(v)/m$, and its section orders divided by j are $v(s)/m$. Applied to the basis with initials Γ_j , the definition of f_j therefore gives

$$Q_j \geq N_j f_j e^{-jA(v)/(mt)} \quad \text{for all sufficiently large } j. \quad (5.19)$$

This comparison occurs at x and does not require the chosen point y to satisfy specialization in degree j . Only the degree- m variation used (5.4).

Combining (5.19) with (5.7), we have

$$\frac{Q_{j_i}}{Q_m} \geq \frac{N_{j_i}}{N_m} \frac{f_{j_i}}{f_m} \exp\left(\left(1 - \frac{j_i}{m}\right) \frac{A(v)}{t}\right). \quad (5.20)$$

For each fixed $i \in \{1, \dots, l-1\}$, the Hilbert asymptotic gives

$$\frac{N_{j_i}}{N_m} \longrightarrow (i/l)^n.$$

Also $j_i \rightarrow \infty$, $f_m \rightarrow c$, and $c = \liminf_j f_j > 0$, so

$$\liminf_{m \rightarrow \infty} \frac{f_{j_i}}{f_m} \geq 1.$$

This does not require the degrees j_i themselves to belong to the subsequence realizing the liminf. Finally, $A(v) \geq a_*$ and $t < n+1$ imply

$$\exp\left(\left(1 - \frac{j_i}{m}\right) \frac{A(v)}{t}\right) \geq \exp\left(\left(1 - \frac{j_i}{m}\right) \frac{a_*}{n+1}\right).$$

Data	Dependence and order of choice
k, C_d, H_d, C'_d	Depend only on (X, L) and d after fixing a very ample multiple kL ; independent of the center, its degree and singularities, and the test.
t_0, a_*, a^*	Fixed by Lemma 4.2 at the hypothetical base point x ; uniform in all sufficiently large degrees and $t \in [t_0, n+1]$.
t , then m	First choose one $t < n+1$ using a_* ; then take a subsequence on which $f_{m,t}(x)$ approaches its positive lower limit. Centers and models may vary with m .
ϵ ; then l	Differentiate at $\epsilon = 0$ for each fixed finite basis. In the partition estimate, let $m \rightarrow \infty$ at fixed l , and only afterwards let $l \rightarrow \infty$.

Table 1: Uniform quantities and the order of limits in the center argument.

All summands in (5.18) are nonnegative for large m , and their number is fixed. Taking the liminf therefore gives

$$\liminf_{m \rightarrow \infty} T_m \geq \frac{d}{l} \sum_{i=1}^{l-1} (i/l)^n \exp\left(\frac{(1-i/l)a_*}{n+1}\right). \quad (5.21)$$

Only now let l tend to infinity. The right side tends to dI_* , which is strictly greater than d/t by (5.17). This contradicts (5.10).

There can consequently be only finitely many degrees on the selected subsequence admitting a positive-dimensional minimizing monomial center. A nonzero test has a nonempty closed center, and every center under consideration contains x . In all remaining degrees that center must therefore be exactly $\{x\}$, proving Proposition 5.1.

6 Isolation and adjoint lifting

We now use a point-centered minimizing pair to construct an effective rational SNC divisor whose coefficient equals the log discrepancy on a nonempty reduced divisor over x , and is strictly smaller on every other component meeting that divisor. This local configuration will allow Kawamata–Viehweg vanishing to lift a nonzero value in the fiber of $P = K_X + (n+1)L$ at x .

6.1 Rational supporting data

Suppose henceforth, for contradiction, that P is not generated at x . Propositions 4.4 and 5.1 give fixed $t < n+1$ and m for which the infimum $f_{m,t}(x)$ is attained and every minimizing monomial pair has center x . Choose such a pair $(s_1, \dots, s_{N_m}; v)$ on a log resolution $\pi : Y \rightarrow X$ of the divisors $D_i = \text{div}(s_i)$. Include all exceptional divisors in the SNC divisor on Y , and denote its components by E_j . Put

$$a_j = A(E_j), \quad z_j = \left(\frac{\text{ord}_{E_j}(D_i)}{a_j} \right)_{i=1}^{N_m}.$$

These vectors are rational and nonnegative, and a_j is a positive integer because X is smooth.

For every component Z of an SNC stratum whose closure has image containing x , let $J(Z)$ be its set of crossing components and form the polytope

$$C_Z = \text{conv}\{z_j : j \in J(Z)\}. \quad (6.1)$$

Only nonempty sets $J(Z)$ are used. There are finitely many such polytopes. A convex representation $z = \sum_j \lambda_j z_j$ is realized by the monomial weights λ_j/a_j ; their discrepancy is one and their section-order vector is z . They can be rescaled to any positive discrepancy. If some weights vanish, the center enlarges to the appropriate stratum closure containing Z . Its image still contains x .

Write

$$a = A(v) > 0, \quad q = \left(\frac{v(s_i)}{a} \right)_{i=1}^{N_m}.$$

The point q belongs to the union of the polytopes (6.1). It belongs to none indexed by a stratum with positive-dimensional image. Indeed, a representation in such a polytope, rescaled to discrepancy a , would give the same objective value and a center containing that positive-dimensional image. This would contradict Proposition 5.1.

Varying the scale of v differentiates the objective at its minimum. With

$$b_i = \frac{t}{m} \frac{\exp(-v(s_i)/m)}{\sum_{h=1}^{N_m} \exp(-v(s_h)/m)}, \quad (6.2)$$

this derivative gives

$$b \cdot q = 1, \quad b_i > 0, \quad m \sum_i b_i = t < n + 1. \quad (6.3)$$

For any C_Z containing q , vary the normalized profile along the segment from q to $q' \in C_Z$, holding the discrepancy equal to a . The one-sided derivative at the minimum is nonnegative. Since the objective decreases with each section order, this says

$$b \cdot q' \leq 1 \quad (q' \in C_Z, q \in C_Z). \quad (6.4)$$

The coefficients needed for vanishing must be rational. We must approximate both q and b while preserving these supporting inequalities and the exclusion of every polytope with positive-dimensional image. The following elementary lemma preserves exactly which indexed polytopes contain the profile; the family may include repeated polytopes.

Lemma 6.1 (Rational supporting data). *Let $\mathcal{C}$ be a finite family of rational polytopes in $\mathbb{R}^N$, and let q belong to their union. Suppose that a vector $b \in \mathbb{R}_{>0}^N$ satisfies*

$$b \cdot q = 1, \quad b \cdot z \leq 1 \quad \text{for every } z \in C \text{ whenever } q \in C \in \mathcal{C}.$$

Fix an open neighborhood V of b . There are rational vectors q^0 arbitrarily close to q and $b^0 \in V \cap \mathbb{Q}_{>0}^N$ such that

1. $q^0 \in C$ if and only if $q \in C$, for each $C \in \mathcal{C}$;
2. $b^0 \cdot q^0 = 1$, and $b^0 \cdot z \leq 1$ on every $C \in \mathcal{C}$ containing q^0 .

Proof. For each polytope containing q , let F_C be its smallest face containing q . The point q lies in the relative interior of F_C . Since the functional b attains its maximum at q , it has value one throughout F_C . The intersection

$$F = \bigcap_{C \ni q} F_C$$

is a nonempty rational polytope, with q in its relative interior. Its rational points are dense in it. Choose a rational point q^0 of F sufficiently close to q to avoid every member of $\mathcal{C}$ not containing q . This is possible because those members form a finite collection of closed sets disjoint from q . The incidence assertion follows, and $b \cdot q^0 = 1$.

For this fixed rational q^0 , the conditions on a functional c are the rational linear equation $c \cdot q^0 = 1$ and the finitely many rational linear inequalities $c \cdot z \leq 1$ at the vertices of the polytopes containing q^0 . Their feasible polyhedron contains b . Rational points are dense in the smallest face of this rational polyhedron containing b . A sufficiently close such point belongs to V and has strictly positive coordinates; take it as b^0 . $\square$

Apply Lemma 6.1, taking the neighborhood of b small enough to preserve the strict budget in (6.3). We obtain rational nonnegative q^0 and rational positive b^0 with

$$b^0 \cdot q^0 = 1, \quad m \sum_i b_i^0 < n + 1, \quad b^0 \cdot q' \leq 1 \quad (q' \in C_Z, q^0 \in C_Z). \quad (6.5)$$

The profile q^0 belongs to no polytope with positive-dimensional image. In particular, $q^0 \neq 0$.

A rational point in a rational convex hull has a rational convex representation. The corresponding rational monomial weights, after clearing denominators, determine a primitive rational ray in the SNC crossing. Toroidal extraction of this ray gives a prime divisor F_0 whose normalized section-order vector is q^0 . The monomial discrepancy formula gives equality in the retraction inequality of Lemma 2.2. Its center is x , by the exclusion of all positive-dimensional-image polytopes. We may retain this divisor on every subsequent common resolution: its orders, discrepancy, and equality under retraction to Y depend on the divisorial valuation, not on the higher model exhibiting it.

6.2 An isolated discrepancy-equality divisor

Let $I = \{i : q_i^0 > 0\}$, and choose an integer $M > 0$ such that M/q_i^0 is an integer for every $i \in I$. Define the effective ideal

$$\mathfrak{c} = \sum_{i \in I} \mathcal{O}_X(-(M/q_i^0)D_i). \quad (6.6)$$

For any prime F , the order of this ideal divided by M is the minimum of the ratios $\text{ord}_F(D_i)/q_i^0$, for $i \in I$. At F_0 all these ratios equal $A(F_0)$. A small contribution from this minimum will preserve equality at F_0 . In the coefficient comparison below, equality in the perturbed bound will force these ratios to agree; positivity of every b_i^0 will also control the coordinates outside I .

Choose an integer u large enough that $\mathfrak{c}(uL)$ is globally generated. Take a smooth projective resolution $\rho : T \rightarrow X$ dominating Y , retaining F_0 , and principalizing $\mathfrak{c}$:

$$\mathfrak{c}\mathcal{O}_T = \mathcal{O}_T(-G).$$

Arrange that G , the pullbacks of the D_i , and all exceptional divisors have SNC support. The divisor $u\rho^*L - G$ is globally generated, since it is the quotient of the pulled-back generating system of $\mathfrak{c}(uL)$.

For a sufficiently small rational number $0 < \eta < 1$, put

$$B = (1 - \eta)\rho^* \left(\sum_i b_i^0 D_i \right) + \frac{\eta}{M} G. \quad (6.7)$$

It is an effective rational divisor with SNC support. Since $D_i \sim mL$, we have

$$\begin{aligned} (n+1)\rho^*L - B &\equiv \left(n+1 - (1-\eta)m \sum_i b_i^0 - \frac{\eta u}{M} \right) \rho^*L \\ &\quad + \frac{\eta}{M} (u\rho^*L - G). \end{aligned} \quad (6.8)$$

The first coefficient is positive when η is sufficiently small, by (6.5). Thus this difference is nef and big.

For a prime divisor F on T , write

$$a_F = A(F), \quad r_F = \sum_j \text{ord}_F(E_j) a_j \leq a_F.$$

Here orders of divisors on Y mean orders of their pullbacks. The number r_F is the discrepancy of the SNC retraction to Y . Let S be the reduced sum of the primes in the chosen SNC support satisfying all three conditions

$$x \in \rho(F), \quad \left(\frac{\text{ord}_F(D_i)}{a_F} \right)_i = q^0, \quad r_F = a_F. \quad (6.9)$$

The retained divisor F_0 is one of them, so $S \neq 0$.

Lemma 6.2. *Every component of S maps to x and has coefficient a_F in B . Every other prime in the SNC support meeting S has coefficient strictly less than its log discrepancy.*

Proof. If F is a component of S , its retraction has normalized profile q^0 , and the image of the retraction center contains $\rho(F)$, hence contains x . That image cannot be positive-dimensional by our choice of q^0 . Consequently $\rho(F) = \{x\}$.

Orders of a sum of ideals are minima of orders. The definition of $\mathfrak{c}$ therefore gives, for every prime F on T ,

$$\frac{\text{ord}_F(G)}{M} = \min_{i \in I} \frac{\text{ord}_F(D_i)}{q_i^0}. \quad (6.10)$$

For a component of S , this is a_F . Together with $b^0 \cdot q^0 = 1$, Equation (6.7) gives $\text{coeff}_F B = a_F$.

Now let F meet a component F_S of S , and choose a point of intersection. Its image y on Y lies over x . Take the stratum through y defined by all SNC components through y , and the corresponding polytope C from (6.1). Every component with positive order along either F or F_S contains the corresponding generic center on Y , and hence contains y . Both retraction profiles therefore belong to C , whenever their retractions are nonzero. In particular $q^0 \in C$, so the supporting inequality in (6.5) applies on this common polytope.

If $r_F = 0$, Lemma 2.2 gives zero order on every D_i , and (6.10) gives zero order on G . Thus $\text{coeff}_F B = 0 < a_F$. If $r_F > 0$, write

$$q' = \left(\frac{\text{ord}_F(D_i)}{r_F} \right)_i \in C, \quad \mu = \min_{i \in I} \frac{q'_i}{q_i^0}.$$

The positive numbers $b_i^0 q_i^0$, for $i \in I$, sum to one. Consequently

$$\mu \leq \sum_{i \in I} b_i^0 q'_i \leq b^0 \cdot q' \leq 1. \quad (6.11)$$

Using (6.10), the exact coefficient ratio is

$$\frac{\text{coeff}_F B}{a_F} = \frac{r_F}{a_F} ((1 - \eta) b^0 \cdot q' + \eta \mu) \leq 1. \quad (6.12)$$

Equality forces $r_F = a_F$ and, since both coefficients in the convex combination are positive, $b^0 \cdot q' = \mu = 1$. The equalities in (6.11) then force $q'_i / q_i^0 = 1$ for every $i \in I$. Strict positivity of all b_i^0 also forces $q'_i = 0$ for $i \notin I$. Thus $q' = q^0$, including its zero coordinates. Because F meets F_S over x , its image contains x . It now satisfies all conditions in (6.9) and is itself a component of S . This proves the strict inequality for every other component meeting S . $\square$

6.3 Lifting and descent

The remaining argument follows the discrepancy-and-vanishing strategy in [9, §1(1.6)], where strict discrepancy inequalities along the divisors meeting an equality divisor allow a nonzero fiber value to lift and descend. The construction above supplies that local configuration on the reduced divisor S .

Define the integral Cartier divisor

$$D = K_T + (n + 1)\rho^* L - \lfloor B \rfloor. \quad (6.13)$$

Its difference from the canonical divisor with fractional boundary is

$$D - (K_T + \{B\}) = (n + 1)\rho^* L - B.$$

The fractional part $\{B\}$ is an effective rational SNC boundary with all coefficients in $[0, 1)$, and the right side is nef and big by (6.8). Theorem 2.4 therefore gives $H^1(T, \mathcal{O}_T(D)) = 0$. The exact sequence of the reduced Cartier divisor S gives a surjection

$$H^0(T, \mathcal{O}_T(D + S)) \longrightarrow H^0(S, \mathcal{O}_S(D + S)). \quad (6.14)$$

Compare the divisor upstairs with the desired adjoint bundle:

$$C := D + S - \rho^*P = K_{T/X} - \lfloor B \rfloor + S. \quad (6.15)$$

On a component F of S , Lemma 6.2 and the integrality of a_F give

$$\text{coeff}_F C = (a_F - 1) - a_F + 1 = 0.$$

For every other prime meeting S , the same lemma gives $\lfloor \text{coeff}_F B \rfloor \leq a_F - 1$, and hence $\text{coeff}_F C \geq 0$. Primes outside the chosen SNC support contribute zero. Thus C is effective on a neighborhood of S , with no component of S in its support. Negative components of C may occur elsewhere, but their supports are disjoint from S ; removing those finitely many supports gives the asserted neighborhood.

Since S is reduced and each of its components maps to x , the restriction $\rho|_S$ factors scheme-theoretically through the point x . Indeed, the pullback of a function vanishing at x vanishes on every component of S , and therefore vanishes on the reduced scheme. It follows that

$$\rho^*P|_S \simeq P|_x \otimes_{\mathbb{C}} \mathcal{O}_S.$$

Choose a nonzero element of the one-dimensional fiber $P|_x$. Multiply the corresponding constant section on S by the canonical rational section of $\mathcal{O}_T(C)$. This latter section is regular near S , because C is effective there, and is nonzero at the generic point of each component of S . The product is therefore a global section of $\mathcal{O}_S(D + S)$, nonzero at those generic points. Notice that no global effectiveness of C is required for this restriction construction.

Lift this section using Equation (6.14). The positive part of C is exceptional over X : on a nonexceptional prime not belonging to S , the coefficient of C is $-\lfloor \text{coeff}_F B \rfloor \leq 0$, while on a component of S it is zero. Since S is nonempty and every component maps to x , any such component, together with the imposed generic nonvanishing, satisfies the hypotheses of Lemma 2.5. The lift therefore descends to a section $s \in H^0(X, P)$ with $s(x) \neq 0$.

The argument also includes dimension one. In that case a component of S can be the nonexceptional point x itself; its coefficient in C is still zero, and the same restriction and descent argument applies.

We have contradicted the assumed failure of generation at x . Since a failure of global generation would occur at a closed point, this proves Theorem 1.1.

6.4 All adjoint powers

Corollary 6.3. *Under the hypotheses of Theorem 1.1, $K_X + mL$ is globally generated for every integer $m \geq n + 1$.*

Proof. Set $r = m - n - 1$. Apply Theorem 1.1 to $X \times \mathbb{P}^r$ and the ample bundle $L \boxtimes \mathcal{O}_{\mathbb{P}^r}(1)$. The adjoint bundle in that theorem is

$$(K_X + mL) \boxtimes \mathcal{O}_{\mathbb{P}^r}(n).$$

Its restriction to $X \times \{z\}$ is globally generated and is isomorphic to $K_X + mL$. For $r = 0$, this is Theorem 1.1 itself. $\square$

References

- [1] Urban Angehrn and Yum-Tong Siu. Effective freeness and point separation for adjoint bundles. *Inventiones Mathematicae*, 122(2):291–308, 1995. doi: 10.1007/BF01231446. URL https://people.math.harvard.edu/~siu/siu_reprints/siu_angehrn_invent1995.pdf.

- [2] Harold Blum and Mattias Jonsson. Thresholds, valuations, and K -stability. *Advances in Mathematics*, 365:107062, 2020. doi: 10.1016/j.aim.2020.107062.
- [3] Sébastien Boucksom and Huayi Chen. Okounkov bodies of filtered linear series. *Compositio Mathematica*, 147(4):1205–1229, 2011. doi: 10.1112/S0010437X11005355.
- [4] Tsz On Mario Chan. A solution to Fujita’s freeness conjecture via an extension theorem with analytic adjoint ideal sheaves, 2024. URL <https://arxiv.org/abs/2411.07129>. Preprint, arXiv:2411.07129v1; withdrawn in v2, 24 November 2024.
- [5] Jean-Pierre Demailly and János Kollár. Semi-continuity of complex singularity exponents and Kähler–Einstein metrics on Fano orbifolds. *Annales scientifiques de l’École Normale Supérieure (4)*, 34(4): 525–556, 2001. doi: 10.1016/S0012-9593(01)01069-2. URL <https://arxiv.org/abs/math/9910118v2>.
- [6] Lawrence Ein and Robert Lazarsfeld. Global generation of pluricanonical and adjoint linear series on smooth projective threefolds. *Journal of the American Mathematical Society*, 6(4):875–903, 1993. doi: 10.1090/S0894-0347-1993-1207013-5. URL <https://doi.org/10.1090/S0894-0347-1993-1207013-5>.
- [7] Osamu Fujino. Kawamata–Viehweg vanishing theorem. Author’s lecture note, 2009. URL <https://www.math.kyoto-u.ac.jp/~fujino/kawamata--viehweg2.pdf>. Version 1.04, 22 July 2009.
- [8] Takao Fujita. On polarized manifolds whose adjoint bundles are not semipositive. In Tadao Oda, editor, *Algebraic Geometry, Sendai, 1985*, volume 10 of *Advanced Studies in Pure Mathematics*, pages 167–178. North-Holland, Amsterdam, 1987. doi: 10.2969/aspm/01010167.
- [9] Takao Fujita. Remarks on Ein–Lazarsfeld criterion of spannedness of adjoint bundles of polarized threefolds, 1993. URL <https://arxiv.org/abs/alg-geom/9311013v1>.
- [10] Richard J. Gardner. The Brunn–Minkowski inequality. *Bulletin of the American Mathematical Society (N.S.)*, 39(3):355–405, 2002. doi: 10.1090/S0273-0979-02-00941-2.
- [11] Luca Ghidelli and Justin Lacini. Logarithmic bounds on Fujita’s conjecture. *Proceedings of the London Mathematical Society (3)*, 128(3):e12591, 2024. doi: 10.1112/plms.12591. URL <https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/plms.12591>. 28 pp.
- [12] Jingjun Han. A linear bound for Fujita’s freeness conjecture, 2026. URL <https://arxiv.org/abs/2609.01574v2>. Preprint, arXiv:2609.01574v2, 11 September 2026.
- [13] Gordon Heier. Effective freeness of adjoint line bundles. *Documenta Mathematica*, 7:31–42, 2002. doi: 10.4171/DM/116.
- [14] Stefan Helmke. On Fujita’s conjecture. *Duke Mathematical Journal*, 88(2):201–216, 1997. doi: 10.1215/S0012-7094-97-08807-4.
- [15] Stefan Helmke. On global generation of adjoint linear systems. *Mathematische Annalen*, 313(4):635–652, 1999. doi: 10.1007/s002080050276.
- [16] Jason A. Howald. Multiplier ideals of monomial ideals. *Transactions of the American Mathematical Society*, 353(7):2665–2671, 2001. doi: 10.1090/S0002-9947-01-02720-9. URL <https://arxiv.org/abs/math/0003232>.
- [17] Kiumars Kaveh and Askold G. Khovanskii. Newton–Okounkov bodies, semigroups of integral points, graded algebras and intersection theory. *Annals of Mathematics (2)*, 176(2):925–978, 2012. doi: 10.4007/annals.2012.176.2.5.
- [18] Yujiro Kawamata. A generalization of Kodaira–Ramanujam’s vanishing theorem. *Mathematische Annalen*, 261:43–46, 1982. doi: 10.1007/BF01456407.
- [19] Yujiro Kawamata. On Fujita’s freeness conjecture for 3-folds and 4-folds. *Mathematische Annalen*, 308(3):491–505, 1997. doi: 10.1007/s002080050085. URL <https://arxiv.org/abs/alg-geom/9510004v2>.
- [20] Robert Lazarsfeld and Mircea Mustață. Convex bodies associated to linear series. *Annales scientifiques de l’École Normale Supérieure (4)*, 42(5):783–835, 2009. doi: 10.24033/asens.2109.
- [21] Andrei Okounkov. Why would multiplicities be log-concave? In Christian Duval, Valentin Ovsienko, and Laurent Guieu, editors, *The Orbit Method in Geometry and Physics*, volume 213 of *Progress in Mathematics*, pages 329–347. Birkhäuser, Boston, MA, 2003. doi: 10.1007/978-1-4612-0029-1_14. URL <https://arxiv.org/abs/math/0002085>.

- [22] Igor Reider. Vector bundles of rank 2 and linear systems on algebraic surfaces. *Annals of Mathematics* (2), 127(2):309–316, 1988. doi: 10.2307/2007055. URL <https://annals.math.princeton.edu/1988/127-2/p04>.
- [23] Eckart Viehweg. Vanishing theorems. *Journal für die reine und angewandte Mathematik*, 335:1–8, 1982. doi: 10.1515/crll.1982.335.1.
- [24] Jarosław Włodarczyk. Simple Hironaka resolution in characteristic zero. *Journal of the American Mathematical Society*, 18(4):779–822, 2005. doi: 10.1090/S0894-0347-05-00493-5.
- [25] Fei Ye and Zhixian Zhu. On Fujita’s freeness conjecture in dimension 5. *Advances in Mathematics*, 371:107210, 2020. doi: 10.1016/j.aim.2020.107210. URL <https://arxiv.org/abs/1511.09154v1>.